HO THE PHONG

BACKEND DEVELOPER | JAVA · SPRING BOOT · NESTJS

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SUMMARY

Software Engineering student at FPT University (degree pending, 2026) with hands-on backend experience in Java (Spring Boot) and TypeScript (NestJS), plus full-stack project work. Built a capstone post-operative monitoring system with Thanh Nhan Hospital. Seeking a Backend Developer role. English: intermediate.

SKILLS

Languages	Java, TypeScript, JavaScript
Backend	NestJS, Spring Boot, Node.js, REST APIs
Databases	MySQL, PostgreSQL, MSSQL, Redis
Frontend	React, TailwindCSS, Flutter
Tools	Git, Docker, Firebase (FCM), TypeORM

PROJECTS

POMS – Post-Operative Monitoring System (Capstone, team of 5). GitHub
NestJS, PostgreSQL, Redis, Docker, React, Flutter

- Backend-focused full-stack developer, working with Thanh Nhan Hospital's ERAS research team on post-operative patient monitoring (Flutter mobile app, React admin dashboard).
- Backend (NestJS/PostgreSQL): database design, clinical triage with scheduled assessments and alert routing, symptom survey, nurse management and assessment-history APIs, role-based access, E2E tests.
- Async & real-time: Redis/Bull queues, cron jobs (daily POD progression), WebSocket real-time alerts, recurring reminders and auto-lock on critical alerts.
- Frontend: set up the Flutter and React base projects; contributed to the Head Nurse web dashboard.

Loop – Real-time community chat app (desktop, full-stack). GitHub
TypeScript, Electron, React, Fastify, Socket.IO, Prisma, PostgreSQL, Redis, LiveKit, Docker

- Backend (Fastify, Prisma/PostgreSQL): REST plus Socket.IO real-time events; Redis for multi-instance scaling, presence and rate limiting; bitmask roles with per-channel permission overrides.
- Security: argon2, short-lived JWT with rotating refresh tokens (reuse detection), signed-URL S3 uploads verified server-side, hardened Electron (sandbox, CSP).
- Desktop client (Electron/React/Vite): LiveKit voice/video with permissions revoked on role change; Vitest/Playwright tests, GitHub Actions CI, Docker Compose deployment, macOS/Windows installers.

CompassEd – AI-powered online learning platform (full-stack). GitHub
Java, Spring Boot, MySQL, Flask, JavaScript, OpenAI API, VNPay, Docker

- Backend (Spring Boot/MySQL): REST APIs for JWT/Google OAuth auth, user management, placement tests, learning roadmaps, and mini/final tests with auto-unlock and level promotion.
- Frontend (Flask, JavaScript): web UI with OpenAI-powered roadmaps and VNPay (sandbox) payment.

EXPERIENCE

Developer - SMART LAB | FPT University *2024 - 2025*

- Developed and tested web application features using Java and Spring Boot in a collaborative team.
- Participated in building a USB-based exam environment (.NET) for FPT University final exams: a sandboxed firmware/virtual machine running EOS (FPT's online exam software) in isolation to prevent cheating.

EDUCATION

FPT University, Hanoi – Bachelor of Software Engineering (degree pending) *2021 - 2026*